Rojae Mighty

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Professional Summary

Physics undergraduate with research experience in scientific computing, machine learning, statistical analysis, and high-energy nuclear physics. Skilled in Python, C++, ROOT, Linux shell scripting, scikit-learn, Monte Carlo simulation, and reproducible data workflows. Interested in graduate research in nuclear physics, quantitative finance, and data-focused technical roles.

Education

Stony Brook University, Stony Brook, NY | **B.S. in Physics**, Expected May 2028 | GPA: 3.7
Relevant Coursework: Electromagnetic Theory, Statistical Mechanics, Computational Methods, Calculus, Linear ALgebra

Research Experience

Brookhaven National Laboratory - Upton, NY

DOE Science Undergraduate Laboratory Internship Researcher | June 2025-August 2025; June 2026-August 2026

- Developed ROOT, Python, and Linux scripts to support electron-ion collision research on 10 million simulated events.
- Built reproducible analysis workflows for studying particle behavior and nuclear effects after high-energy collisions.
- Worked independently within an 8-person research team, translating physics requirements into validated code and organized outputs.

Center for Frontiers in Nuclear Science (CFNS) - Stony Brook, NY

Undergraduate Researcher, Nuclear and High-Energy Physics | January 2025-Present

- Built Python and scikit-learn pipelines to reconstruct key kinematic quantities from 10 million particle-collision events.
- Used C++, ROOT, Monte Carlo event generators, and numerical simulations to study gluon saturation and geometric scaling at the Electron-Ion Collider.
- Created reusable workflows that improved repeatability, verification, and sharing of analysis results across research tasks.
- Presented research at DIS, APS Global Physics Summit, Yale URC, NSBP, APS DNP, and Stony Brook Physics Colloquium.

Projects

Market Microstructure Forecasting

GitHub: https://github.com/iRojae/market-microstructure-forecasting | January 2025-Present

- Built a Python machine-learning project with pandas, NumPy, and scikit-learn to forecast short-horizon price movement from limit-order-book data.
- Engineered trading features including order-flow imbalance and bid-ask spread changes; evaluated models with walk-forward validation and transaction-cost-aware backtesting.

J.P. Morgan Quantitative Research Virtual Experience Program - Forage

January 2026

- Completed 4 quantitative research tasks covering commodity analysis, derivatives pricing, credit-risk modeling, and FICO score grouping.
- Modeled natural gas prices using 48 monthly observations and built a storage-contract pricer for a 1,000,000 MMBtu sample contract valued at $583,118 net.
- Built probability-of-default models on 10,000 borrower records using 6 financial features; evaluated a 7,500/2,500 train-test split with AUC = 1.0000 and Brier score = 0.0031.

Honors and Presentations

- 1st Place for Physical Sciences Presentation at Yale Undergraduate Research Conference, January 2026
- Outstanding Oral Presentation Award, National Society of Black Physicists Conference, January 2025
- Barry Barish Undergraduate Research Travel Award and Edward Guiliano Global Fellowship, Stony Brook University, January 2026
- Research presentations: DIS, APS Global Physics Summit, APS DNP, Yale URC, NSBP, and Stony Brook Physics Colloquium

Technical Skills

Programming: Python, C++, C, Unix/Linux shell scripting
Data and Machine Learning: pandas, NumPy, scikit-learn, feature engineering, statistical analysis, walk-forward validation, dynamic programming
Research and Tools: ROOT, Monte Carlo event generators, numerical simulation, Git, GitHub, LaTeX, Linux/Unix environments
Domains: Electron-Ion Collider physics, particle-collision data, gluon saturation, geometric scaling, market microstructure